

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY (ICDFA)



SBT-DF203 LAB 4 REPORT: SMTP Email Traffic Forensics

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Course Title: Basic Networking Skills for Digital Forensics

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Case/Lab Identifier: SBT-DF203-Lab4-AbubakariSadiqueHamidu

Date: 15 September 2026

1. Source Materials and Slide Alignment

This investigation aligns directly with the SBT-DF203 lecture series covering Email Traffic Forensics.

The practical analysis examines the supplied smtp.pcap capture and focuses on an authentic SMTP email transfer. The investigation was conducted to determine the communication timeline, client software, message content, network endpoints, TCP ports, and hardware MAC addresses. Authentication exchanges and recovered credentials were handled as confidential forensic evidence and are redacted in this report.

2. Lab Scenario

A historical SMTP packet capture (smtp.pcap) was supplied for forensic evaluation.

The objectives of the investigation were to:

- Reconstruct the email exchange timeline.
- Identify the client and server systems.
- Reassemble SMTP application-layer commands and server response codes.
- Decode Base64 authentication challenges in an offline environment.
- Recover and examine the email message content and MIME structure.
- Identify the mail client software.
- Determine the relevant IP addresses, TCP ports, and MAC addresses.
- Assess the transport security of the SMTP session, including STARTTLS.
- Document the evidential limitations associated with encrypted and unencrypted SMTP traffic.

All analysis was conducted using a preserved working copy of the evidence inside an isolated Kali Linux Docker container.

3. Learning Outcomes Demonstrated

The investigation demonstrated the following learning outcomes:

- Explained the role of SMTP and distinguished between plaintext SMTP, STARTTLS, and implicit TLS.
- Identified SMTP commands, server response codes, and authentication exchanges.
- Reassembled an SMTP TCP stream and reconstructed message headers and body parts.
- Decoded Base64 values in a controlled offline environment using Python.
- Extracted timestamps, client software, IP addresses, TCP ports, and MAC addresses.
- Assessed evidential limitations where encryption is used or where network captures contain unencrypted application-layer traffic.

4. Legal, Ethical and Safety Rules Compliance

The investigation was conducted in accordance with the following legal, ethical, and forensic safety requirements:

- Analysis was executed strictly within an authorised, isolated virtual laboratory container (sbtdf203-lab).
- The analysis environment was completely isolated from production and public networks.
- Original capture files were preserved and all analysis was performed using a hash-verified working copy.
- Recovered usernames, passwords, and message contents were treated as confidential assessment evidence.
- Sensitive authentication information has been redacted from this public-facing report.

5. Tools and Environment Used

Item	Tool Used	Forensic Purpose
Operating System	Kali Linux (kalilinux/kali-rolling container)	Environment hosting the forensic toolkit

Item	Tool Used	Forensic Purpose
Packet Analysis	Wireshark and TShark v4.6.6	Filtering, frame dissection, and TCP stream reassembly
Decoder	Python 3 base64 module	Offline decoding of Base64 authentication strings
Hashing	sha256sum	Evidence integrity and chain-of-custody verification
Evidence	smtp.pcap	Historical training traffic capture

6. Lab Folder Structure and Evidence Preparation

A standard forensic workspace hierarchy was established before analysis:

SBT-DF203-Lab4/

- |— evidence/ # Preserved original capture files
- |— working/ # Working copies for active investigation
- |— exported/ # Exported objects and payloads
- |— reports/ # TSV extractions, hashes, and text summaries
- |— screenshots/ # Contemporaneous terminal evidence
- |— scripts/ # Automation and Python decoding scripts

The original capture was stored as:

evidence/smtp.pcap

A working copy was then created as:

working/smtp_working.pcap

The working copy was created using `cp --preserve=timestamps`, preserving the original file

timestamps.

The SHA-256 hash of both files was verified as follows:

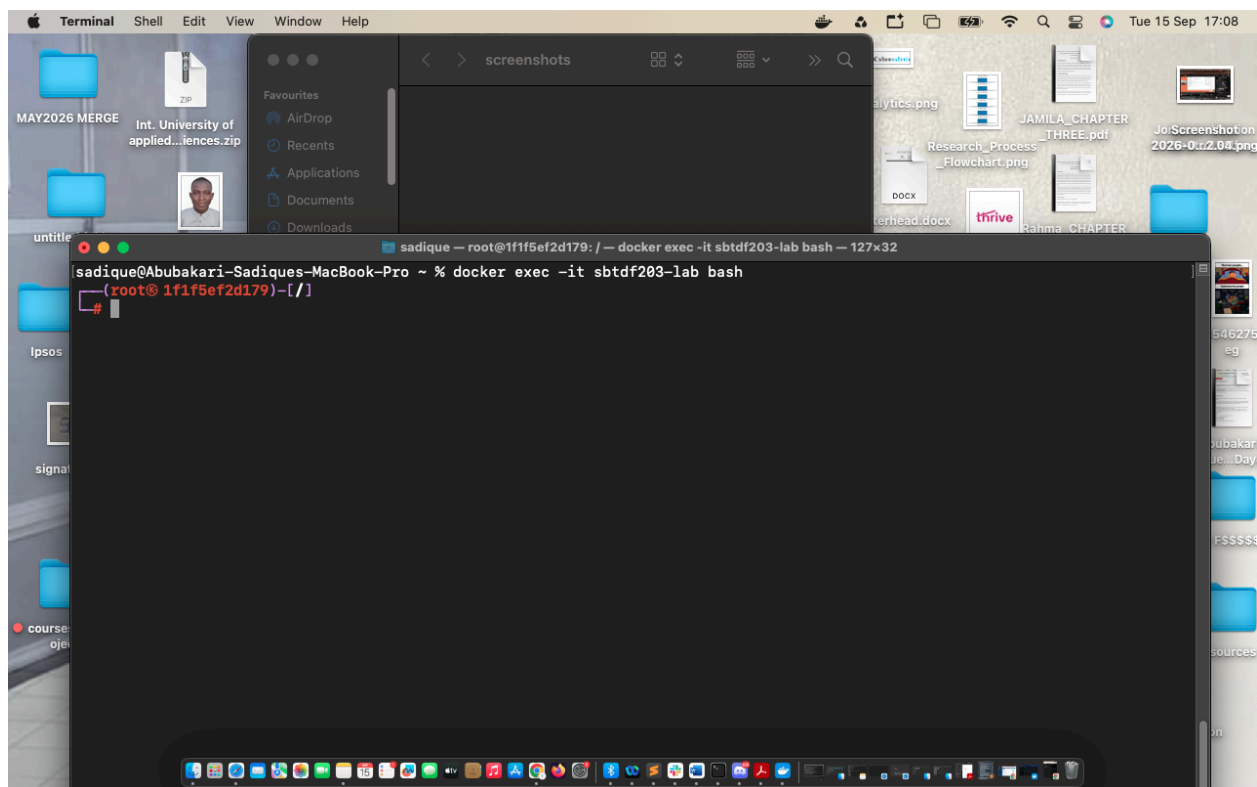
```
396827c13483df92d713c8e4125a2dfdf8309c6454792ecf635688d5e8ff313d
```

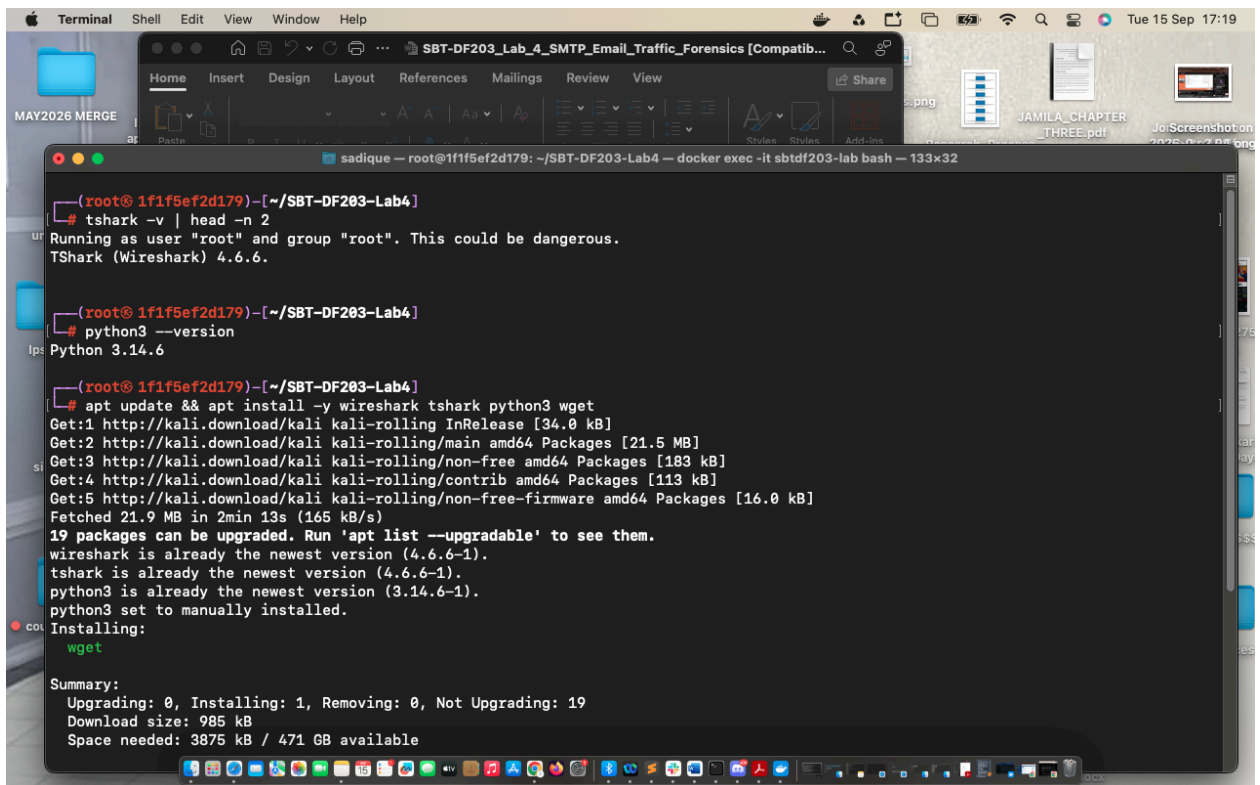
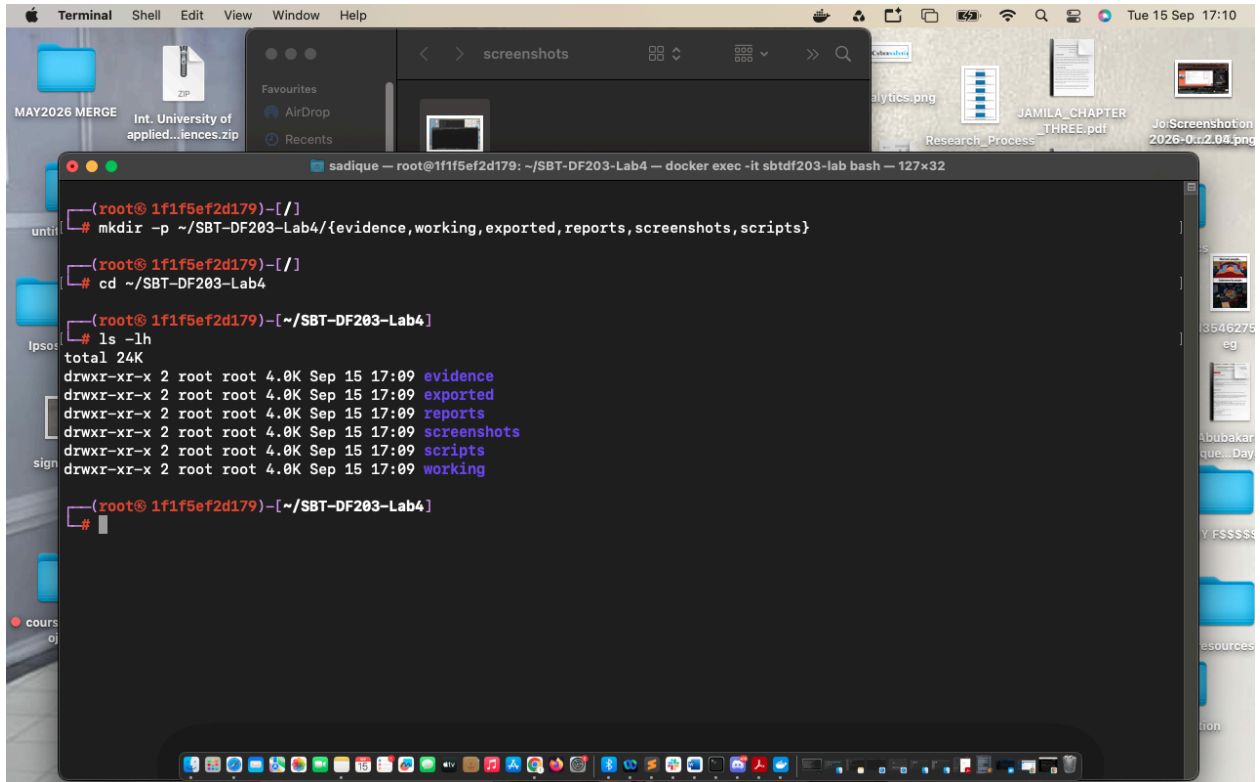
```
evidence/smtp.pcap
```

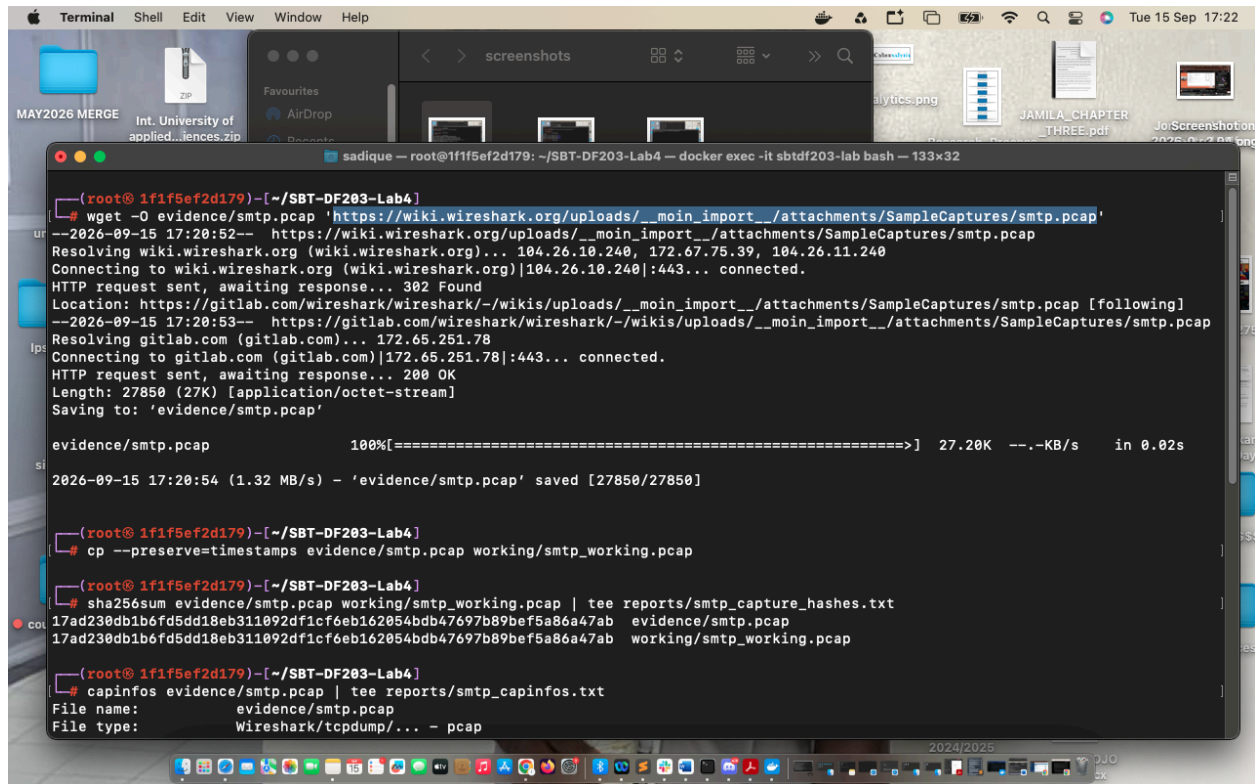
```
396827c13483df92d713c8e4125a2dfdf8309c6454792ecf635688d5e8ff313d
```

```
working/smtp_working.pcap
```

The matching SHA-256 hashes confirm that the working copy remained unchanged from the original capture.







7. Evidence and Chain-of-Custody Worksheet

Field	Student / Forensic Entry
Case/Lab Identifier	SBT-DF203-Lab4-AbubakariSadiqueHamidu
Trainee Name	Abubakari-Sadique Hamidu
Date and Time Started	2026-09-15 17:00 WAT
Evidence File Name(s)	smtp.pcap / smtp_working.pcap
Source / Generation Method	Authorised ICDFA Historical Dataset (Wireshark Sample Captures)
Original SHA-256	396827c13483df92d713c8e4125a2dfdf8309c

Field	Student / Forensic Entry
	6454792ecf635688d5e8ff313d
Working-Copy SHA-256	396827c13483df92d713c8e4125a2dfdf8309c 6454792ecf635688d5e8ff313d
Analysis Workstation/VM	macOS Host / Kali Linux Docker Container (sbtdf203-lab)
Notes on Changes	Working copy created using cp --preserve=timestamps; no byte-level changes identified

8. Part A – Inventory the Capture and Locate SMTP Streams

The TCP conversation summary (reports/tcp_conversations.txt) and SMTP frame inventory (reports/smtp_packet_inventory.tsv) were used to establish the principal endpoints and capture characteristics.

8.1 Client System

- **IP Address:** 10.10.1.4
- **TCP Port:** 1470 (ephemeral client port)
- **MAC Address:** 00:e0:1c:3c:17:c2

8.2 SMTP Server

- **IP Address:** 74.53.140.153
- **TCP Port:** 25
- **MAC Address:** 00:1f:33:d9:81:60

8.3 Capture Characteristics

-
- The image shows a macOS desktop with several open windows. In the foreground, a terminal window titled "sadique ~ root@1f1f5ef2d179: ~/SBT-DF203-Lab4 — docker exec -it sbtfd203-lab bash — 133x32" is active. Inside the terminal, the user has run the command:
- ```
(root@1f1f5ef2d179) [~/SBT-DF203-Lab4]
tshark -r working/smtp_working.pcap -q -z conv,tcp | tee reports/tcp_conversations.txt
```
- This command captures TCP conversations from the file "smtp\_working.pcap". The terminal output shows the following details:
- Running as user "root" and group "root". This could be dangerous.
- TCP Conversations  
Filter:<No Filter>
- | Duration                 |                      | <-            | ->           | Total        | Relative    | D |
|--------------------------|----------------------|---------------|--------------|--------------|-------------|---|
|                          |                      | Frames Bytes  | Frames Bytes | Frames Bytes | Start       |   |
| 10.10.1.4:1470<br>7.5777 | <-> 74.53.140.153:25 | 25 1980 bytes | 28 22 kB     | 53 24 kB     | 0.036986000 |   |
- Below the table, the terminal shows the raw packet capture data for the selected conversation:
- ```
(root@1f1f5ef2d179) [~/SBT-DF203-Lab4]
# tshark -r working/smtp_working.pcap -Y 'smtp' -T fields \
-e frame.number -e frame.time -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e _ws.col.Info \
> | tee reports/smtp_packet_inventory.tsv
```
- Running as user "root" and group "root". This could be dangerous.
- ```
6 2009-10-05T06:06:08.219663000+0000 74.53.140.153 25 10.10.1.4 1470 S: 220-xc90.websitewelcome.com ESMTP
Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500 | We do not authorize the use of this system to transport unsolicited, | and/or bulk e
-mail.
7 2009-10-05T06:06:08.224809000+0000 10.10.1.4 1470 74.53.140.153 25 C: EHLO GP
GP [122.162.143.157] | SIZE 52428800 | PIPELINING | AUTH PLAIN LOGIN | STARTTLS | HELP S: 250-xc90.websitewelcome.com Hello
10 2009-10-05T06:06:08.568729000+0000 10.10.1.4 1470 74.53.140.153 25 C: AUTH LOGIN
11 2009-10-05T06:06:08.711081000+0000 74.53.140.153 25 10.10.1.4 1470 S: 334 VXNlcmShbWU6
12 2009-10-05T06:06:08.911655000+0000 10.10.1.4 1470 74.53.140.153 25 C: User: Z3VycGFydGFWQHBhdHJpb3RzMlmlu
13 2009-10-05T06:06:09.253544000+0000 74.53.140.153 25 10.10.1.4 1470 S: 334 UGFzc3dvcmQ6
14 2009-10-05T06:06:09.254118000+0000 10.10.1.4 1470 74.53.140.153 25 C: Pass: cHVuamFiQDEyMw==
15 2009-10-05T06:06:09.613798000+0000 74.53.140.153 25 10.10.1.4 1470 S: 235 Authentication succeeded
16 2009-10-05T06:06:09.614414000+0000 10.10.1.4 1470 74.53.140.153 25 C: MAIL FROM: <gurpartap@patriots.in>
```

The SMTP exchange was reconstructed from the packet capture. The following key commands and response codes were identified:

| Sequence | Direction       | Command / Code | Meaning       | Timestamp (UTC) |
|----------|-----------------|----------------|---------------|-----------------|
| 1        | Server → Client | 220            | Service Ready | 2009-10-05      |

| Sequence | Direction       | Command / Code | Meaning                                                       | Timestamp (UTC)               |
|----------|-----------------|----------------|---------------------------------------------------------------|-------------------------------|
|          |                 |                | (xc90.websitewelcome.com Exim 4.69)                           | 06:06:08.219663               |
| 2        | Client → Server | EHLO           | Extended SMTP client greeting (EHLO GP)                       | 2009-10-05<br>06:06:08.224809 |
| 3        | Client → Server | AUTH LOGIN     | Initiates SMTP login authentication exchange                  | 2009-10-05<br>06:06:08.568729 |
| 4        | Client → Server | MAIL FROM      | Envelope sender declaration (<gurpartap@patriots.in>)         | 2009-10-05<br>06:06:09.614414 |
| 5        | Client → Server | RCPT TO        | Envelope recipient declaration (<raj_deol2002in@yahoo.co.in>) | 2009-10-05<br>06:06:09.957250 |
| 6        | Client → Server | DATA           | Initiates message payload transfer                            | 2009-10-05<br>06:06:10.320203 |
| 7        | Client → Server | QUIT           | Requests session                                              | 2009-10-05                    |

| Sequence | Direction | Command / Code | Meaning     | Timestamp (UTC) |
|----------|-----------|----------------|-------------|-----------------|
|          |           |                | termination | 06:06:14.763825 |

**Evidence Reference:** Screenshot #3 – 03\_smtp\_commands\_responses.png

## 10. Part C – Decode Base64 Authentication Evidence

The AUTH LOGIN mechanism generated two Base64-encoded authentication prompts from the SMTP server.

The values were decoded offline using Python.

### 10.1 Offline Base64 Decoding Results

#### OFFLINE BASE64 DECODING RESULTS

Username Prompt (334): VXNlcm5hbWU6

Decoded: Username:

Captured Username:

[REDACTED]

Password Prompt (334): UGFzc3dvcmQ6

Decoded: Password:

Captured Password:

[REDACTED]

### 10.2 Redaction Protocol

Because the recovered authentication credentials constitute sensitive information, the actual

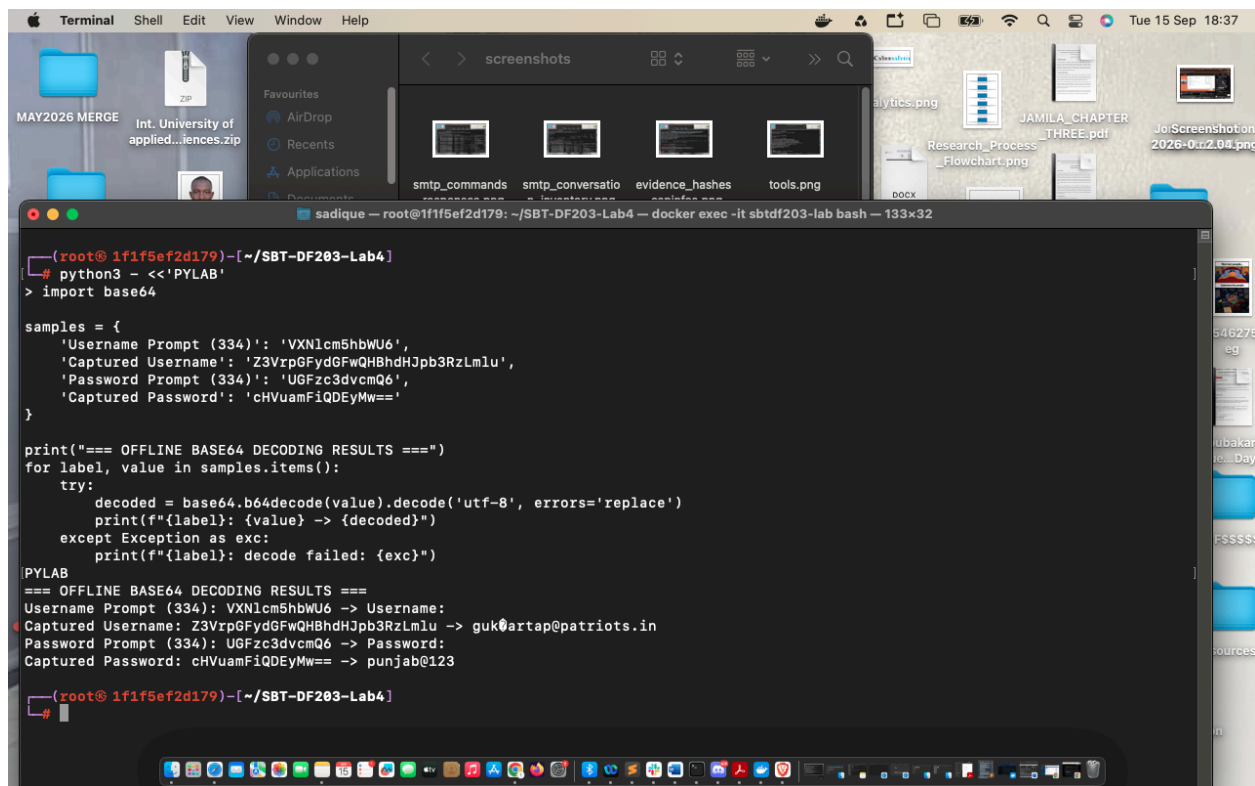
username and password are not reproduced in this report.

The username and password have been redacted in accordance with forensic confidentiality principles.

### 10.3 Forensic Finding

The analysis demonstrates that Base64 is an encoding mechanism rather than an encryption mechanism. Base64-encoded credentials can therefore be recovered when the underlying SMTP traffic is transmitted without encryption.

This illustrates the security risks associated with transmitting authentication information over plaintext SMTP connections.



```
(root@1f1f5ef2d179)~/SBT-DF203-Lab4
python3 - <<'PYLAB'
> import base64

samples = {
 'Username Prompt (334)': 'VXN1cm5hbWU6',
 'Captured Username': 'Z3VrpGFydGFwQHhhdHJpb3RzLmlu',
 'Password Prompt (334)': 'UGFzc3dvcmQ6',
 'Captured Password': 'cHVuamFiQDEyMw=='
}

print("=== OFFLINE BASE64 DECODING RESULTS ===")
for label, value in samples.items():
 try:
 decoded = base64.b64decode(value).decode('utf-8', errors='replace')
 print(f"{label}: {value} -> {decoded}")
 except Exception as exc:
 print(f"{label}: decode failed: {exc}")

PYLAB
=== OFFLINE BASE64 DECODING RESULTS ===
Username Prompt (334): VXN1cm5hbWU6 -> Username:
Captured Username: Z3VrpGFydGFwQHhhdHJpb3RzLmlu -> guk@artapatriots.in
Password Prompt (334): UGFzc3dvcmQ6 -> Password:
Captured Password: cHVuamFiQDEyMw== -> punjab@123

(root@1f1f5ef2d179)~/SBT-DF203-Lab4
#
```

## 11. Part D – Reconstruct the Email Message

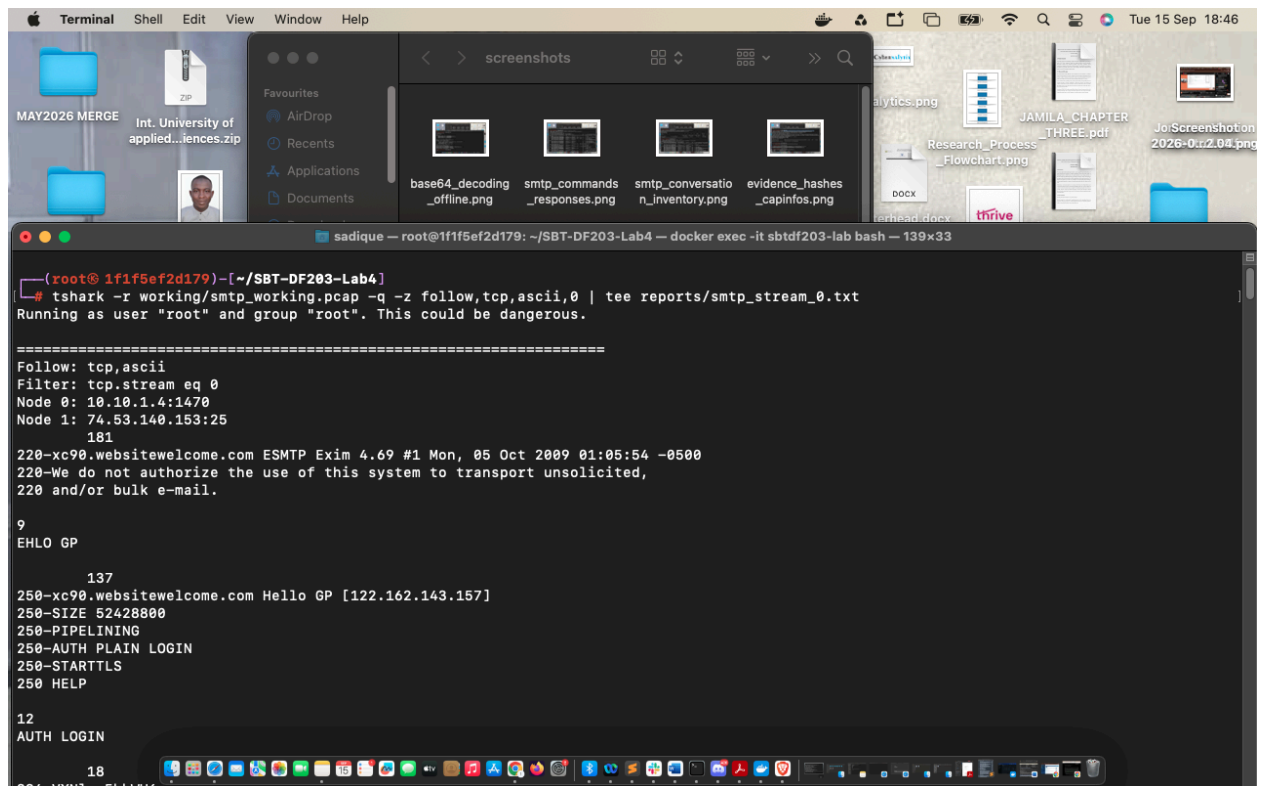
TCP Stream 0 was reassembled using the captured SMTP traffic. The reconstruction consolidated 14 segmented TCP DATA fragments into the logical RFC-formatted email message.

The following message characteristics were identified:

- **Date:** Mon, 5 Oct 2009 11:36:07 +0530

- **From:** "Gurpartap Singh" <gurpartap@patriots.in>
- **To:** <raj\_deol2002in@yahoo.co.in>
- **Subject:** SMTP
- **Message-ID:** <000301ca4581\$ef9e57f0\$cedb07d0\$@in>
- **MIME-Version:** 1.0
- **Content-Type:** multipart/mixed
- **MIME Structure:** multipart/alternative, including text/plain and text/html
- **Client Software (X-Mailer):** Microsoft Office Outlook 12.0

The reconstructed message demonstrated how SMTP application data can be recovered from a packet capture when the session is transmitted without transport encryption.



The screenshot shows a macOS desktop environment. In the background, a Finder window is open, displaying a folder named 'screenshots' which contains several PNG files: 'base64\_decoding\_offline.png', 'smtp\_commands\_responses.png', 'smtp\_conversation\_inventory.png', and 'evidence\_hashes\_capinfos.png'. Other files visible on the desktop include 'MAY2026 MERGE', 'Int. University of applied...lences.zip', 'analytics.png', 'JAMILA\_CHAPTER\_THREE.pdf', 'Research\_Process\_Flowchart.png', 'JoiScreenshoton 2026-0...2.04.png', 'thrive', and 'Jamiladocx'. In the foreground, a terminal window titled 'sadique — root@1f1f5ef2d179: ~/SBT-DF203-Lab4 — docker exec -it sbtdf203-lab bash — 139x33' is open. The terminal shows the execution of a tshark command to follow an SMTP stream and the resulting output, which includes SMTP protocol details such as EHLO, AUTH, and LOGIN commands and responses.

```
(root@ 1f1f5ef2d179)~[~/SBT-DF203-Lab4]
tshark -r working/smtp_working.pcap -q -z follow,tcp,ascii,0 | tee reports/smtp_stream_0.txt
Running as user "root" and group "root". This could be dangerous.

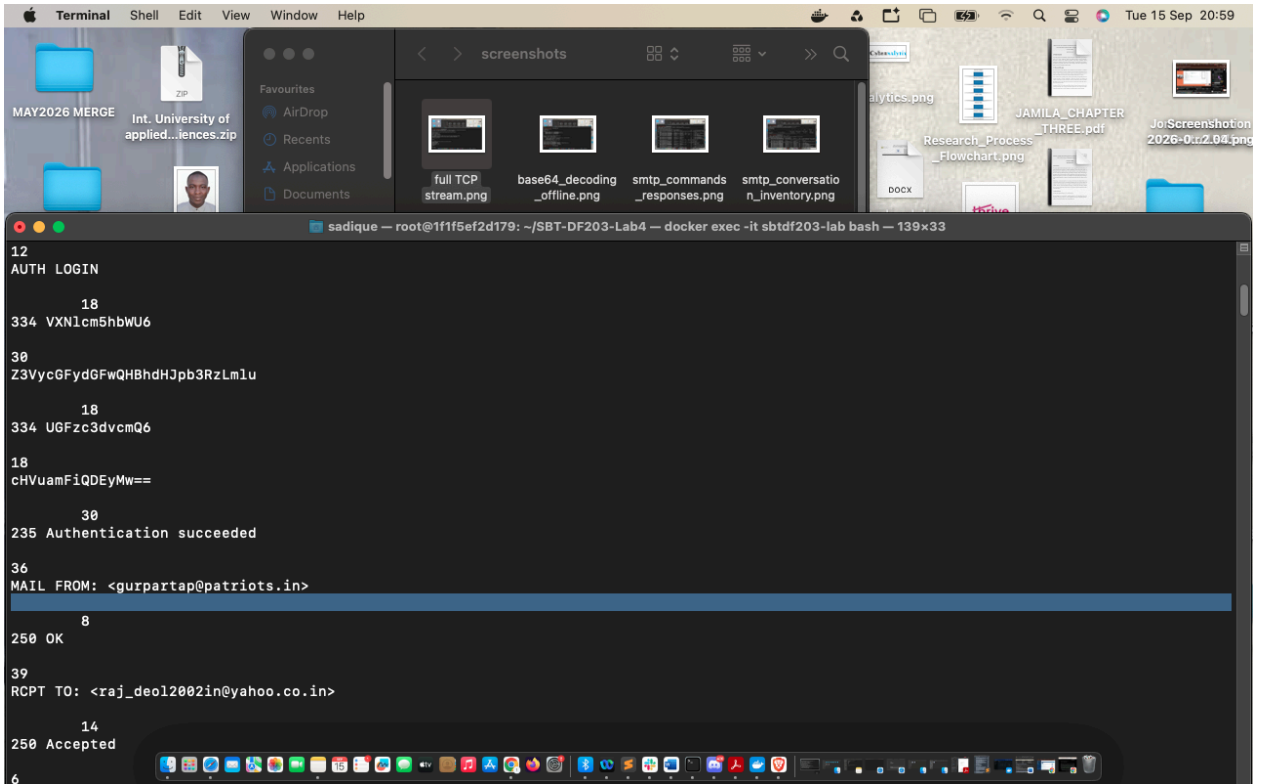
=====
Follow: tcp,ascii
Filter: tcp.stream eq 0
Node 0: 10.10.1.4:1470
Node 1: 74.53.140.153:25
181
220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500
220-We do not authorize the use of this system to transport unsolicited,
220 and/or bulk e-mail.

9
EHLO GP

137
250-xc90.websitewelcome.com Hello GP [122.162.143.157]
250-SIZE 52428800
250-PIPELINING
250-AUTH PLAIN LOGIN
250-STARTTLS
250 HELP

12
AUTH LOGIN

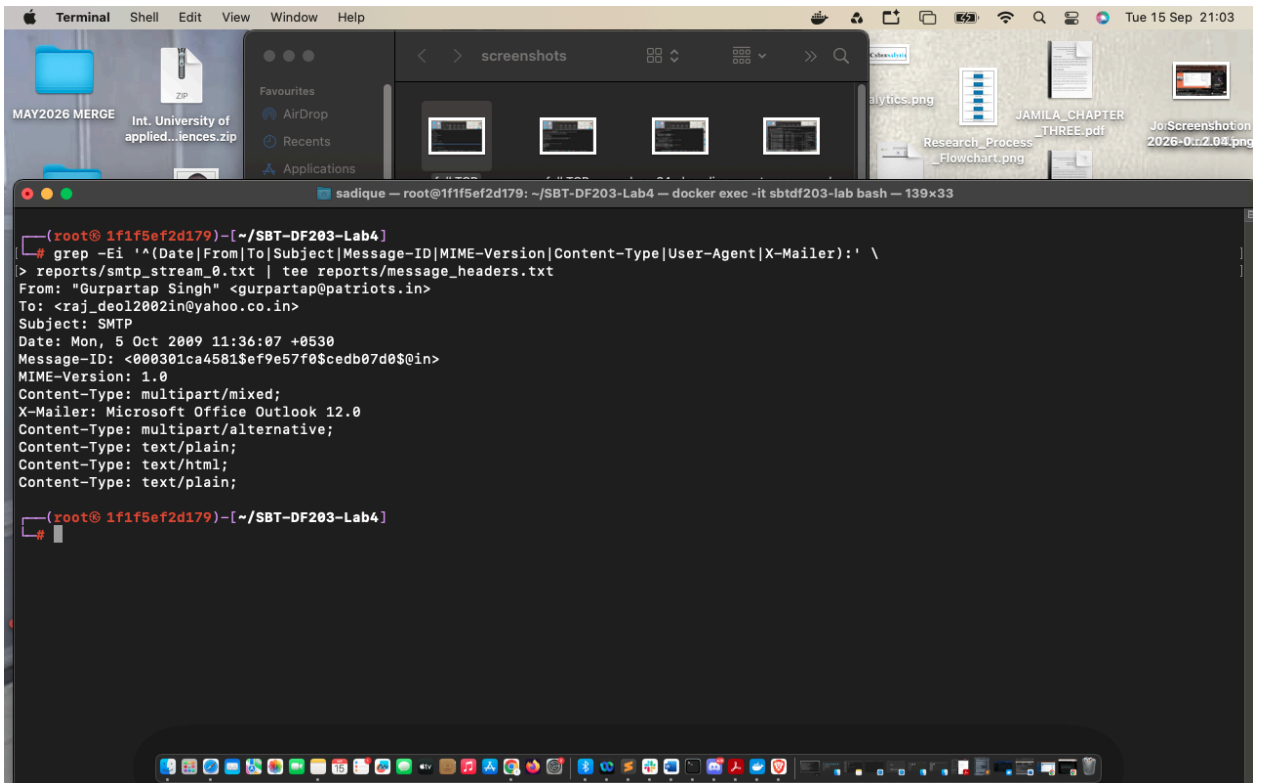
18
334 JGN1cm55bWU6
```



The screenshot shows a macOS desktop environment. A File Explorer window is open, displaying a folder named 'screenshots' containing several image files: 'full TCP stream.png', 'base64\_decoding\_offline.png', 'smtp\_commands\_responses.png', and 'smtp\_conversation\_inventory.png'. In the background, a Terminal window is active, showing an SMTP session. The session starts with an 'AUTH LOGIN' command, followed by base64-encoded credentials. The server responds with '235 Authentication succeeded'. Then, a 'MAIL FROM' command is issued, and the server responds with '250 OK'. Finally, a 'RCPT TO' command is issued, and the server responds with '250 Accepted'.

```
Terminal Shell Edit View Window Help
sadique — root@1f1f5ef2d179: ~/SBT-DF203-Lab4 — docker exec -it sbtdf203-lab bash — 139x33

12 AUTH LOGIN
18
334 VXNlcm5hbWU6
30
Z3VycGFydGFWQHBhdHJpb3RzLmlu
18
334 UGFzc3dvcmQ6
18
cHVuamFiQDEyMw==
30
235 Authentication succeeded
36 MAIL FROM: <gurpartap@patriots.in>
8
250 OK
39 RCPT TO: <raj_deol2002in@yahoo.co.in>
14
250 Accepted
6
```



The screenshot shows a macOS desktop environment. A Terminal window is active, displaying the output of a grep command. The command is used to extract email headers from an SMTP stream. The output shows the 'From' and 'To' fields of an email, along with other header information like 'Subject', 'Date', 'Message-ID', and 'MIME-Version'.

```
Terminal Shell Edit View Window Help
sadique — root@1f1f5ef2d179: ~/SBT-DF203-Lab4 — docker exec -it sbtdf203-lab bash — 139x33

(root@1f1f5ef2d179)~/SBT-DF203-Lab4
grep -Ei '^(Date|From|To|Subject|Message-ID|MIME-Version|Content-Type|User-Agent|X-Mailer):' \
> reports/smtp_stream_0.txt | tee reports/message_headers.txt
From: "Gurpartap Singh" <gurpartap@patriots.in>
To: <raj_deol2002in@yahoo.co.in>
Subject: SMTP
Date: Mon, 5 Oct 2009 11:36:07 +0530
Message-ID: <000301ca4501$ef9e57f0$cedb07d0$@in>
MIME-Version: 1.0
Content-Type: multipart/mixed;
X-Mailer: Microsoft Office Outlook 12.0
Content-Type: multipart/alternative;
Content-Type: text/plain;
Content-Type: text/html;
Content-Type: text/plain;

(root@1f1f5ef2d179)~/SBT-DF203-Lab4
#
```

A screenshot of a macOS desktop environment. At the top, a Chrome browser window is open with the address bar showing 'screenshots'. Below it, a terminal window is active, displaying a series of commands and their outputs. The terminal prompt is '(root@ 1f1f5ef2d179)~[~/SBT-DF203-Lab4]'. The first command is 'grep -Ei '^([Date|From|To|Subject|Message-ID|MIME-Version|Content-Type|User-Agent|X-Mailer]):' \> reports/sntp\_stream\_0.txt | tee reports/message\_headers.txt'. The output shows email headers for a message from 'Gurpartap Singh' to 'raj\_deol2002in@yahoo.co.in', dated 'Mon, 5 Oct 2009 11:36:07 +0530'. The second command is 'sed -E 's/gurpartap@patriots.in/g\*\*\*\*\*@patriots.in/g; s/punjab@123/p\*\*\*\*\*3/g' \> reports/sntp\_stream\_0.txt > reports/reconstructed\_email\_redacted.txt'. The terminal window is titled 'sadique - root@1f1f5ef2d179: ~/SBT-DF203-Lab4 - docker exec -it sbtdf203-lab bash - 139x33'. The macOS dock at the bottom contains various application icons, including Finder, Safari, and several utility apps. The system status bar at the bottom right shows the date 'Tue 15 Sep 21:04'.

## 12. Part E – Determine Client, Hosts and Network Metadata

The following network and application metadata were identified from the packet capture:

### Client System

- **IP Address:** 10.10.1.4
- **MAC Address:** 00:e0:1c:3c:17:c2
- **TCP Port:** 1470

### SMTP Server

- **IP Address:** 74.53.140.153
- **MAC Address:** 00:1f:33:d9:81:60
- **TCP Port:** 25

## Mail Client Software

The email was identified as having been generated using:

### Microsoft Office Outlook 12.0

This was determined from the X-Mailer header contained in the reconstructed email.

```
(root@1f1f5ef2d179) ~/SBT-DF203-Lab4
tshark -r working/smtp_working.pcap -Y 'smtp' -T fields \
> -e frame.number -e frame.time_epoch -e eth.src -e eth.dst -e ip.src -e tcp.srcport \
> -e ip.dst -e tcp.dstport -e tcp.stream \
> | tee reports/smtp_network_metadata.tsv
Running as user "root" and group "root". This could be dangerous.
6 1254722768.219663000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0
7 1254722768.224809000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
9 1254722768.566183000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0
10 1254722768.568729000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
11 1254722768.911081000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0
12 1254722768.911655000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
13 1254722769.253544000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0
14 1254722769.254118000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
15 1254722769.613798000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0
16 1254722769.614414000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
17 1254722769.956765000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0
18 1254722769.957250000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
19 1254722770.319708000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0
20 1254722770.320203000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
21 1254722770.661679000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0
22 1254722770.692743000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
23 1254722770.692786000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
24 1254722770.692804000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
25 1254722770.692823000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
26 1254722770.695115000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.53.140.153 250
28 1254722770.695623000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.53.140.153 250
29 1254722770.696248000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.53.140.153 250
30 1254722770.696634000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.53.140.153 250
32 1254722771.104972000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
33 1254722771.104998000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
35 1254722771.1469759000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
```

## 13. Part F – Encryption and Evidential Limitations

### 13.1 STARTTLS Assessment

In Frame 9, the SMTP server advertised the following capability:

250-STARTTLS

However, the client did not issue a STARTTLS command during the captured session.

Therefore, STARTTLS was **advertised but not negotiated**, leaving the observed SMTP



application-layer exchange unencrypted.

### **13.2 Evidential Limitations**

When SMTP traffic is protected by TLS/SMTPS, application-layer information such as the following may not be directly recoverable from the packet capture:

- Authentication credentials
- SMTP commands and responses
- Email headers
- Message bodies
- MIME content
- Attachments

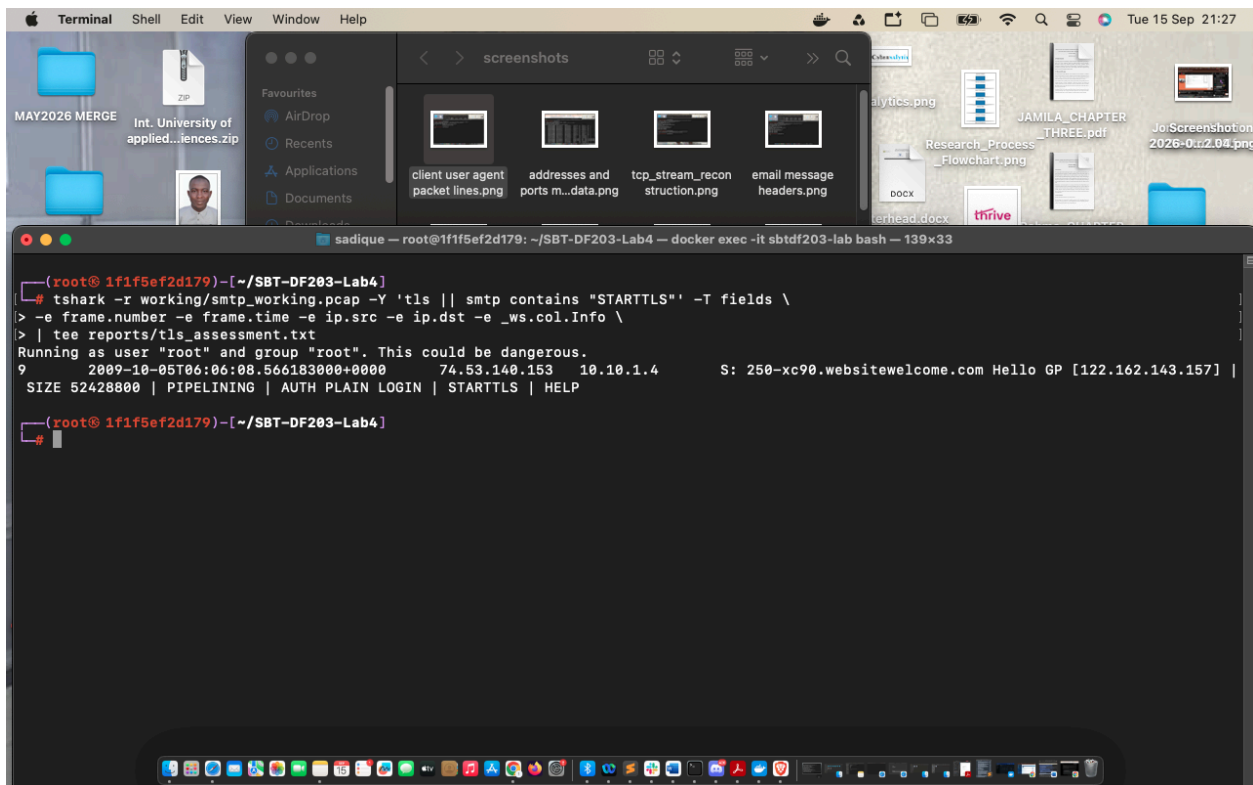
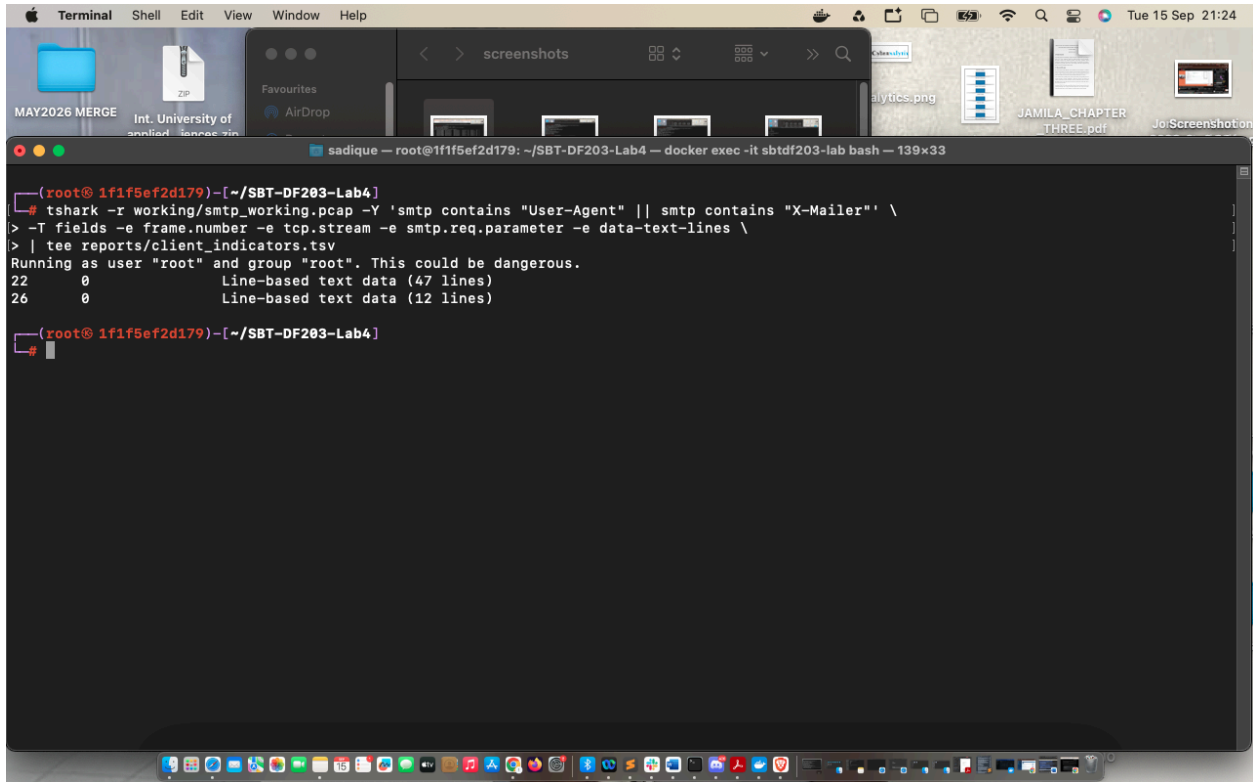
However, lower-layer network metadata may remain available for forensic analysis, including:

- IP addresses
- MAC addresses, where captured
- TCP ports
- Packet timing
- Packet sizes
- Byte counts
- TCP connection information

### **13.3 Port 25 Does Not Automatically Determine Encryption**

The use of TCP port 25 alone does not establish whether an SMTP session is encrypted.

A forensic examiner must inspect the actual protocol exchange to determine whether STARTTLS was advertised and, importantly, whether it was actually negotiated.



## 14. Required Findings Worksheet

| Question / Investigative Field           | Forensic Finding                                                                     |
|------------------------------------------|--------------------------------------------------------------------------------------|
| When did the SMTP session start and end? | 2009-10-05 06:06:08.219663 to<br>06:06:15.105467 UTC — approximately 7.58<br>seconds |
| Client IP/MAC and port                   | IP: 10.10.1.4   MAC: 00:e0:1c:3c:17:c2   Port:<br>1470                               |
| Server IP/MAC and port                   | IP: 74.53.140.153   MAC: 00:1f:33:d9:81:60  <br>Port: 25                             |
| SMTP server banner                       | xc90.websitewelcome.com ESMTP Exim<br>4.69 #1                                        |
| Client software                          | Microsoft Office Outlook 12.0                                                        |
| Authentication method                    | AUTH LOGIN using Base64 encoding                                                     |
| Envelope sender and recipient            | gurpartap@patriots.in →<br>raj_deol2002in@yahoo.co.in                                |
| Message From/To/Subject                  | "Gurpartap Singh" →<br>raj_deol2002in@yahoo.co.in   Subject: SMTP                    |
| Message body type                        | multipart/mixed, containing<br>multipart/alternative, text/plain, and text/html      |
| Attachment present?                      | No binary file attachment identified                                                 |
| STARTTLS/TLS observed?                   | STARTTLS advertised (250-STARTTLS) but<br>not negotiated                             |

| Question / Investigative Field | Forensic Finding                                                                                           |
|--------------------------------|------------------------------------------------------------------------------------------------------------|
| Key security finding           | Base64 provides encoding, not encryption; the captured SMTP session was transmitted without negotiated TLS |

## 15. Overall Forensic Findings

The investigation established that the supplied packet capture contained a complete SMTP email exchange between a client system at 10.10.1.4 and an SMTP server at 74.53.140.153 over TCP port 25.

The SMTP server identified itself as:

```
xc90.websitewelcome.com ESMTP Exim 4.69 #1
```

The client initiated an SMTP session using EHLO, followed by AUTH LOGIN, MAIL FROM, RCPT TO, DATA, and QUIT.

The authentication exchange used Base64 encoding rather than encryption. The server advertised STARTTLS, but the client did not negotiate it. Consequently, the captured SMTP application-layer traffic remained observable in plaintext.

TCP stream reassembly enabled reconstruction of the email headers and MIME structure. The X-Mailer header identified Microsoft Office Outlook 12.0 as the mail client.

The investigation also demonstrated the forensic value of preserving original evidence, verifying SHA-256 hashes, maintaining a separate working copy, documenting the analysis environment, and redacting sensitive authentication information in the final report.

## 16. Academic Integrity and Authorisation Statement

### Student Declaration

I confirm that I completed this laboratory exercise in an authorised environment, preserved the supplied evidence, did not target any third-party system, and accurately documented my own commands, observations, analysis, and conclusions.

**Signed:** Abubakari-Sadique Hamidu

**Student ID:** 2025-FWSD-11392

**Date:** 15 September 2026